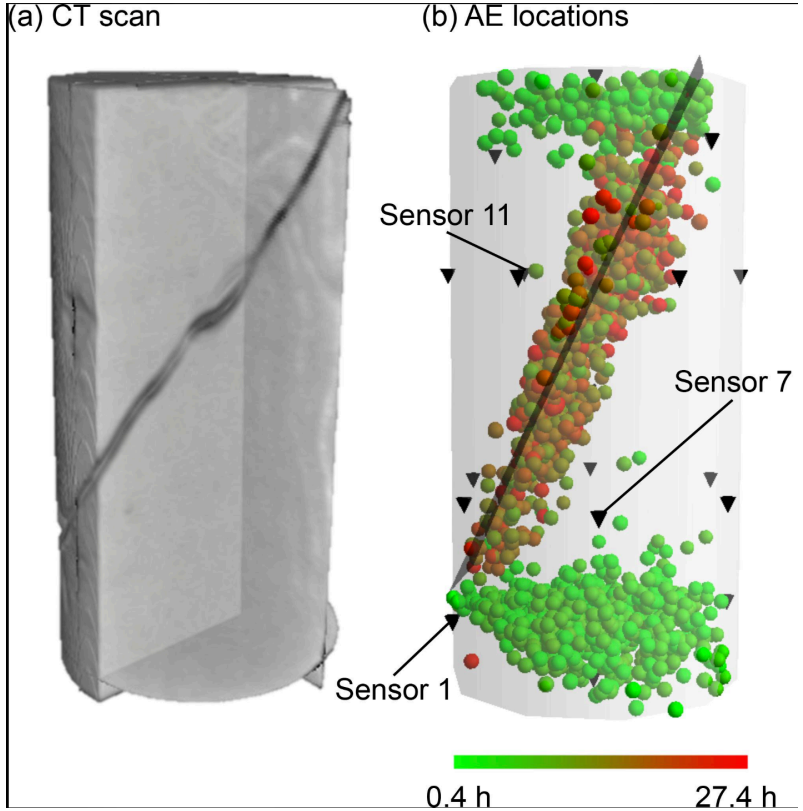
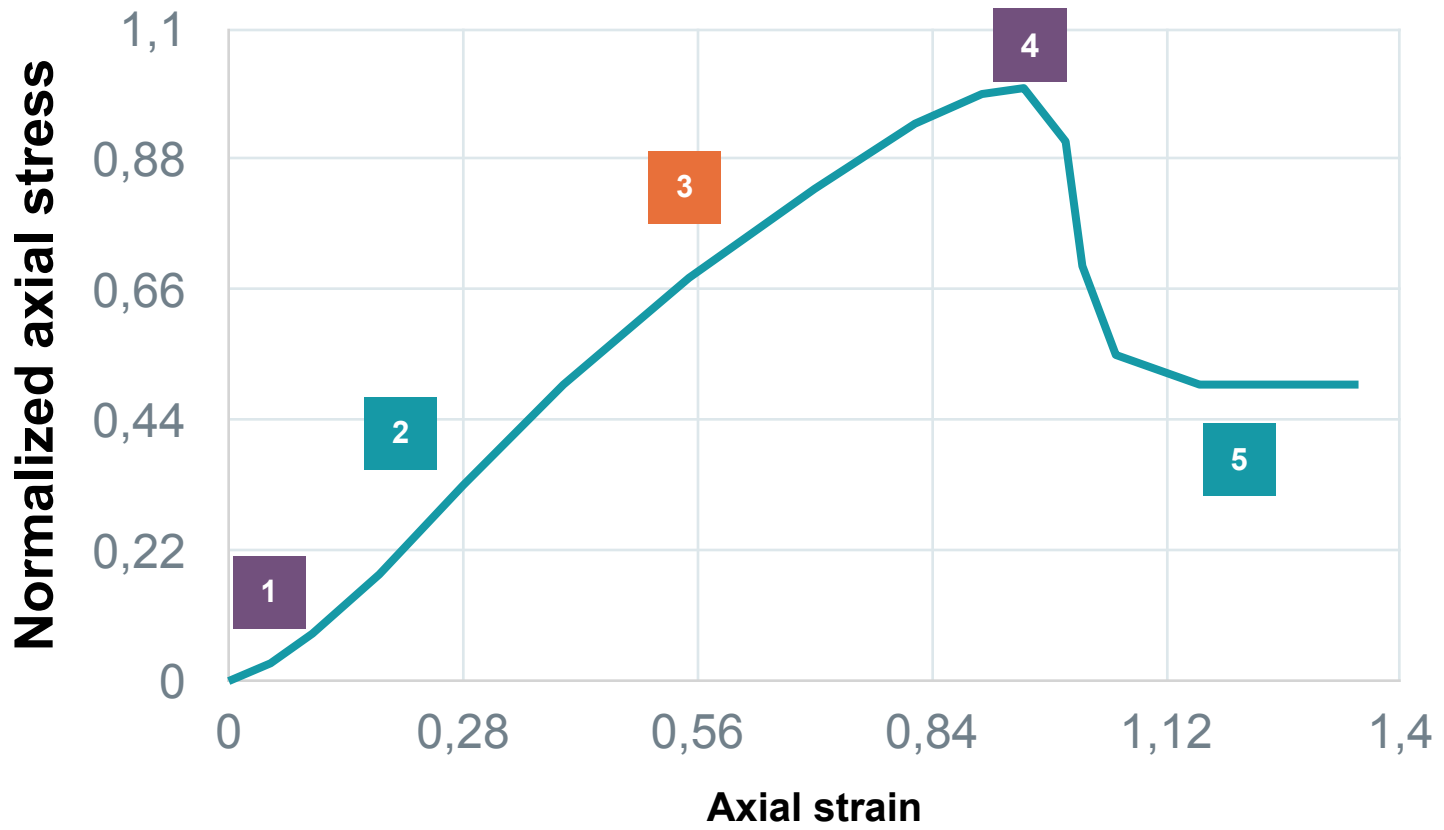
Can we hear a rock approaching failure?

From microcrack initiation to AE detection during a UCS experiment
A.K.A fracture mechanics

One experiment · three levels of observation: stress–strain, cracking, acoustic signals



Read failure as a sequence of physical processes



1 Closure
initial curvature

2 Elastic loading
near-linear response

3 Distributed damage
dilatancy and VP loss

4 Localization
damage concentrates before peak

5 Fault sliding
post-peak residual strength

Peak stress separates strengthening from weakening. It does not mark the onset of damage.

A UCS curve is the macroscopic consequence of many microscopic events.

The key puzzle: failure is sudden, damage is not

Before peak stress

Distributed microcracks nucleate, extend, interact and increasingly concentrate deformation.

At peak / instability

A connected rupture surface develops and the rock rapidly loses load-bearing capacity.

What can we measure?

Mechanical signals capture the sample response; Ultrasonic waveforms internal changes, AE records short-lived elastic waves from rapid local sources.

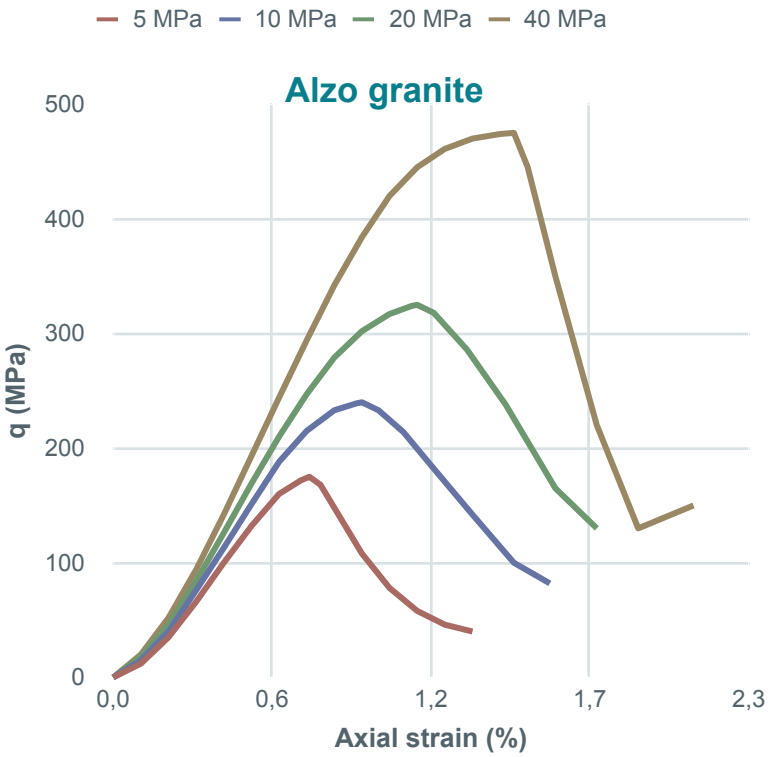
Question: where along the stress–strain curve would you expect AE to begin?

What should you be able to do after this lecture?

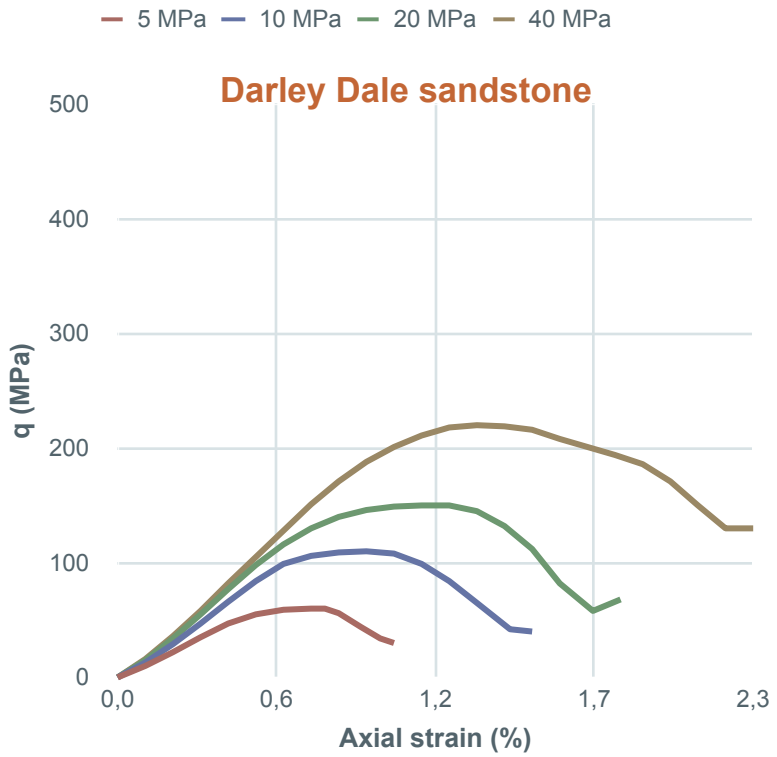
1. Explain why cracks develop under compression
2. Distinguish crack nucleation, propagation, coalescence and localization
3. Explain what generates and what limits an AE measurement
4. Choose a defensible detection threshold and compute AE rate
5. Predict the main AE observations during our UCS laboratory

Why does increasing confining pressure make rocks stronger, and how can this macroscopic observation be explained by the initiation, growth and interaction of microscopic cracks?

Start from an experimental observation, not from a failure criterion.



Sharper weakening after a much higher peak



Lower peaks with extended strain softening

Look at the curves

- Peak differential stress increases.
- The Young Modulus changes
- The post-peak response changes.
- Failure geometry also changes?

What changes microscopically?

Conceptual curves (not experimental data). Compare with Paterson & Wong, brittle-failure chapters.

Why does confining pressure suppress brittle opening?

A macroscopic strength trend points toward the role of local tensile cracks.

Low confinement

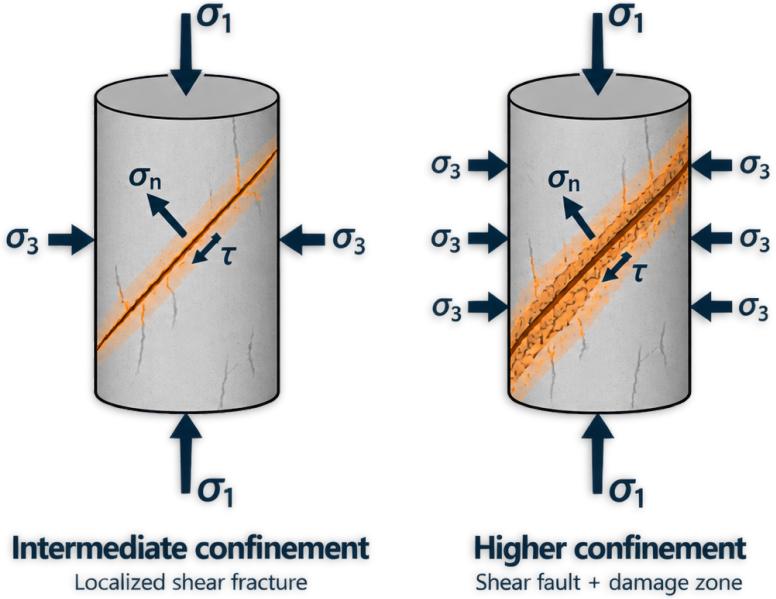
Cracks open more readily; axial splitting and dilatancy can develop.





Higher confinement

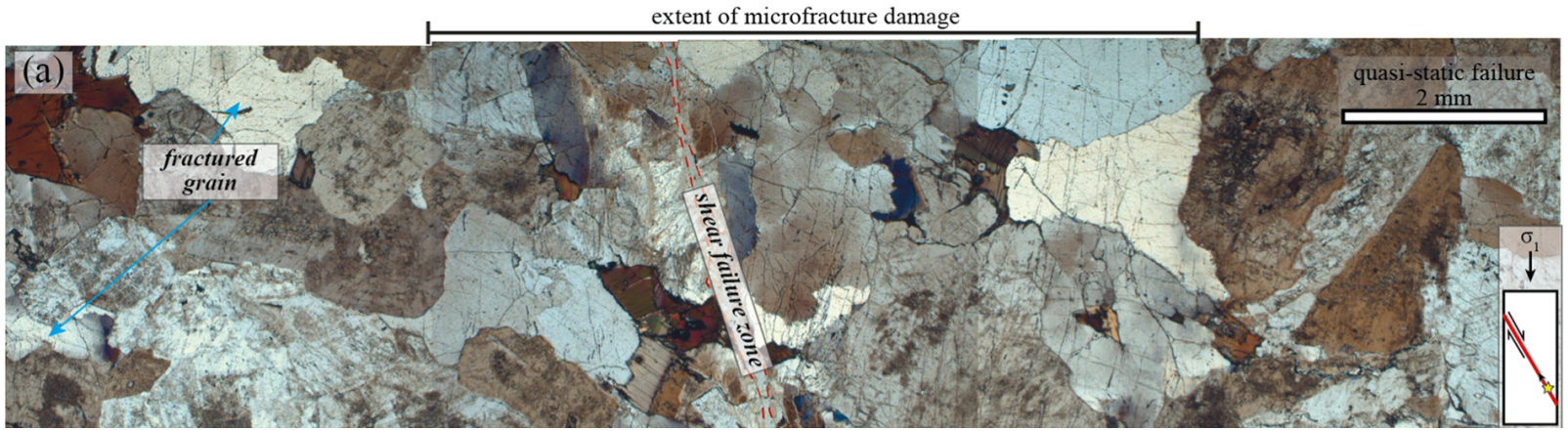
Opening is restrained; greater differential stress is generally needed for distributed cracking and failure.



Same rock ≠ failure mode

Conceptual synthesis: Paterson & Wong, experimental brittle strength and failure modes.

Rock begins as a heterogeneous stress field



Geometric sources

Pre-existing cracks, pores and grain-boundary corners

Material sources

Elastic mismatch between adjacent minerals

Contact sources

Grain contacts, cleavage and weak interfaces

A uniform applied stress produces a strongly nonuniform local stress.

Aben et al. (2020), Fig. 10; Paterson & Wong (2005), Section 6.2.1

Which flaw activates first?

Activity 1

Three closed flaws have equal length but different orientations.

β = 15°

closed flaw

Nearly parallel to σ_1

β = 30°

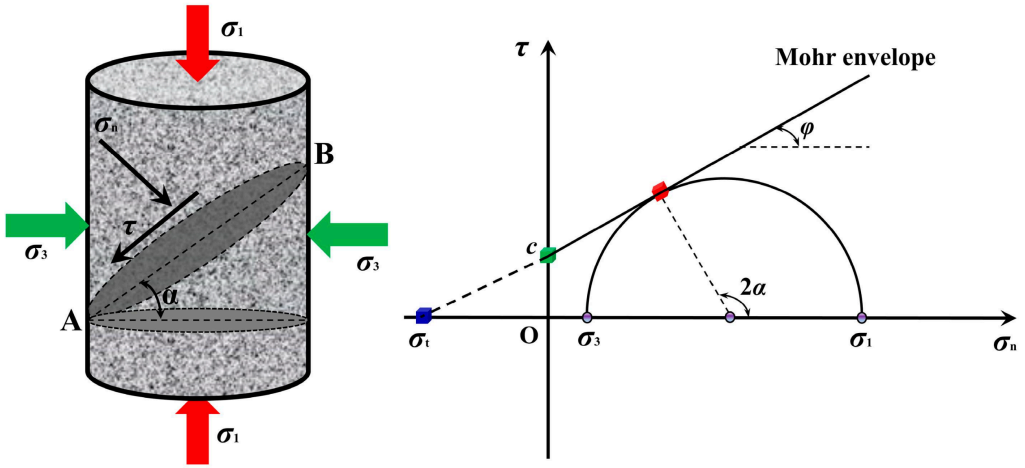
closed flaw

Intermediate angle

β = 75°

closed flaw

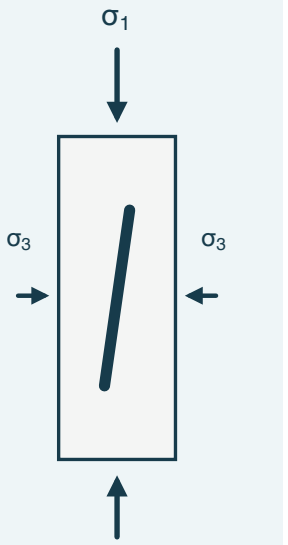
Nearly normal to σ_1



Rank the flaws by likelihood of sliding. State the stress component that controls your ranking.

Flaw orientation controls the resolved shear traction

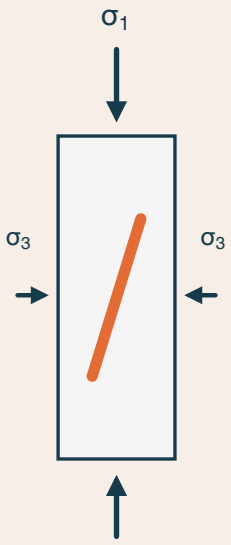
Equal flaw length, identical friction, different angle to σ_1



β = 15 °

Nearly parallel to σ_1

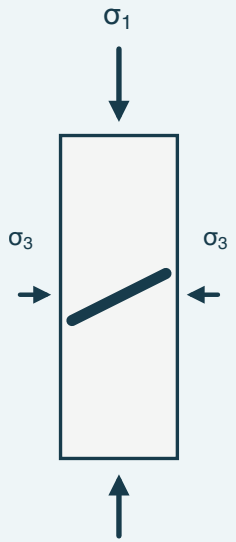
small resolved shear



β = 30 °

Intermediate angle

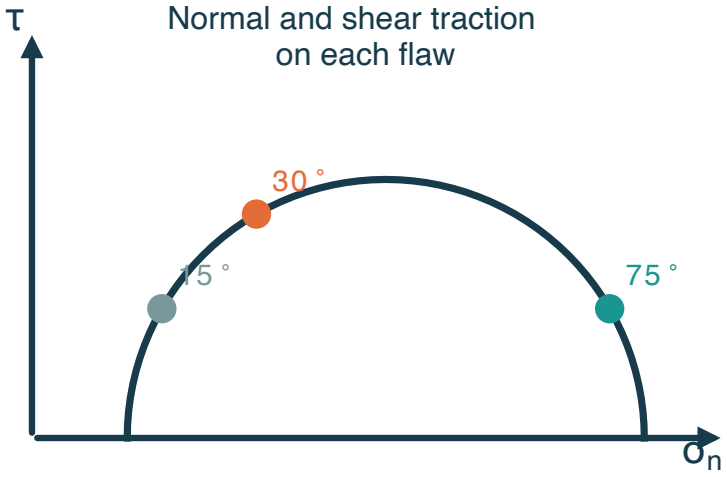
largest resolved shear



β = 75 °

Nearly normal to σ_1

small resolved shear



$$\sigma_n = (\sigma_1 + \sigma_3)/2 - (\sigma_1 - \sigma_3)\cos 2\beta / 2$$

$$\tau = (\sigma_1 - \sigma_3)\sin 2\beta / 2$$

Prediction: the 30° flaw has the largest τ.

Resolved traction relations: Paterson & Wong (2005), Section 6.2.1